

CENG 589 Homework 2

Intrinsic Symmetry Axis Detection

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Abstract

This study investigates the problem of detecting the intrinsic symmetry axis of 3D meshes—an axis defined purely by the internal geometry of a shape, independent of its orientation in space. Two complementary methods are implemented and analyzed: Geodesic Isocurve Intersection, which estimates the axis based on overlapping isocurves from shape extremities, and Pairwise Harmonics Analysis, which leverages harmonic scalar fields under Dirichlet boundary constraints to identify symmetric regions. Key preprocessing steps include geodesic distance computation using Dijkstra’s algorithm, Farthest Point Sampling for identifying extremities, and a similarity metric to select intrinsically symmetric point pairs. Results show that while both approaches perform well on simple and relatively symmetric meshes, they face challenges with complex, non-uniform geometries. Modifications, such as expanding boundary constraints, offer considerable improvements.

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1. Introduction

In this assignment, we aim to identify the intrinsic reflectional symmetry axis of a given 3D model. The intrinsic axis divides the mesh into two parts that are symmetric with respect to the model’s internal geometry, independent of its embedding in 3D space. This is distinct from standard reflectional symmetry, focusing instead on the intrinsic properties of the surface.

To accomplish this, we are asked to implement two complementary approaches:

Geodesic Isocurve Intersection [1]: We begin by detecting the shape’s extremities using Farthest Point Sampling (FPS). From these extremities, we generate geodesic isocurves across the mesh and identify the first intersection where the isocurves sufficiently overlap, providing an initial estimation of the symmetry axis.

Pairwise Harmonics Analysis: As a second approach, we employ pairwise harmonics, following the method described in [2], to robustly detect the intrinsic reflectional symmetry axis.

Each approach involves intermediate tasks, such as computing geodesic distances and establishing meaningful correspondences between points across the mesh. Together, these methods provide a reliable framework for uncovering the underlying intrinsic symmetry of 3D shapes.

2. Initial and Intermediate Tasks

To find the shortest distance between two vertices on the surface, Dijkstra’s shortest path algorithm is implemented, and to find shape extremities, farthest point sampling (FPS) is used. For both Geodesic Isocurve Intersection and Pairwise Harmonics Analysis parts, we need two intrinsically symmetric sample points. However, FPS does not ensure that points are intrinsically symmetric. Therefore, I use the so-called similarity test described in [2]. For similarity test, a function $\Psi(g_i, g_j)$ is defined as

$$\Psi(g_i, g_j) = \frac{1}{n} \sum_{k=1}^n \left| \frac{g_i^k - g_j^k}{g_i^k + g_j^k} \right| \quad (1)$$

where g_i and g_j are the geodesic descriptors of points i and j , defined as the lengths of the geodesic rings propagating from points i and j at equal distances respectively.

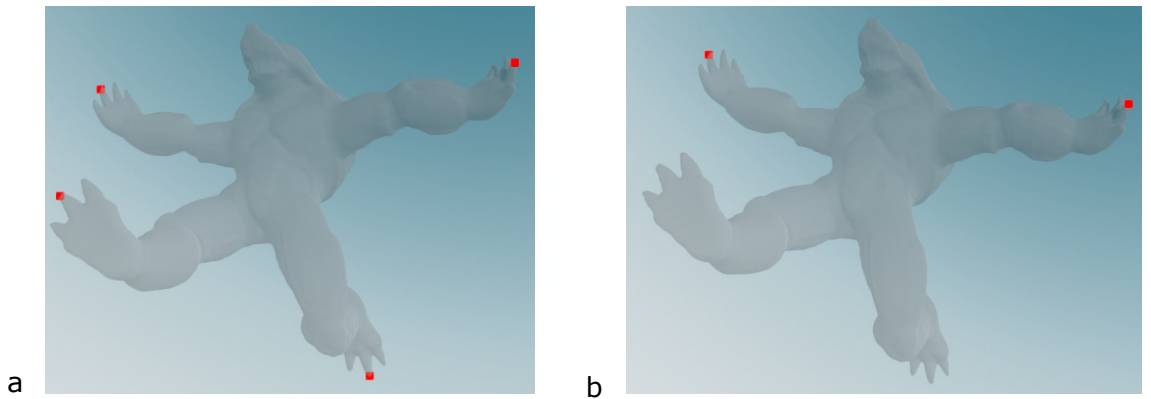


Figure 1: a) FPS points, b) selected pair by similarity test

For two point pairs, the one with the low Ψ value is a better point pair, meaning points of that pair are closer to being intrinsically symmetric. To find the best point

pair, I run FPS for four sample points as depicted. For every pairwise combination of those four points, Ψ values are calculated. The pair with the lowest Ψ value is selected. The FPS points and selected pair for an example mesh is depicted in Figure 1

The algorithm to compute geodesic rings as follows.

- *Initialize*: Create empty dictionaries for ring lengths and vertices.
- *Compute Distances*: Use Dijkstra's algorithm to calculate geodesic distances from FPS points.
- *Determine Rings*: Calculate maximum distance and number of rings.
- *Group Vertices*: For each ring, define bounds (lower bound, upper bound) and find vertices within bounds.
- *Compute Ring Lengths*: Sum edge lengths for vertices in the ring, avoiding double-counting.
- *Store Results*: Save ring lengths and vertices for each ring.
- *Return*: Return ring lengths and ring vertices

Also in [2], the so-called geodesic distance test is used to omit close point pairs; however, since FPS samples farthest points, they are not close to each other, so we do not need to use the geodesic distance test.

In this assignment, I used an SDK called Open3D which is an open-source Python library for 3D data processing and visualization.

3. Geodesic Isocurves

In this part, we generate geodesic isocurves from two extreme points across the mesh and identify the first intersection where the isocurves sufficiently overlap. However, this intersection does not necessarily give the symmetry axis because the distances of an FPS point to the points on the axis of symmetry are not the same for each point. This is more evident if the symmetry axis is relatively larger. Therefore, I use this overlap as an initial estimation of the symmetry axis.

I define the sufficient overlap as if there are at least one overlapped vertex. Those vertices are on the symmetry axis but they are only on a small region of the symmetry axis. Therefore, I needed some other points. I found distances from each extremity to every vertex on the mesh. I extracted the vertices where these distances are close to each other. When, we connect these vertices, we can identify the symmetry axis.

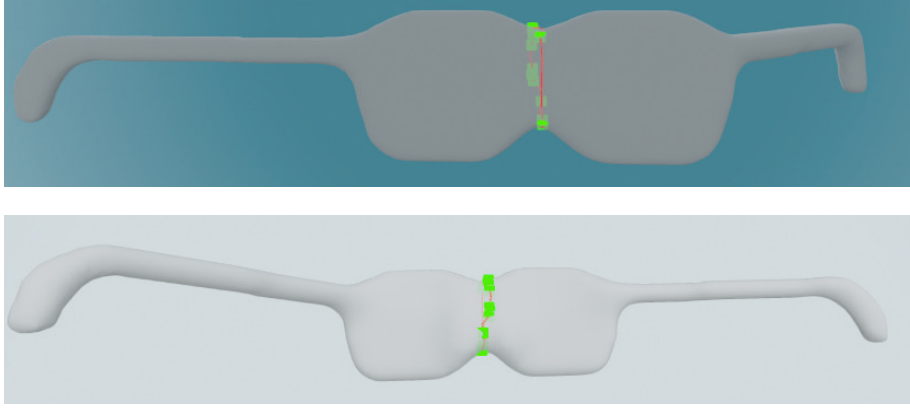


Figure 2: The intrinsic symmetry of the glasses mesh by geodesic isocurve intersection



Figure 3: The intrinsic symmetry of the armadillo mesh by geodesic isocurve intersection

The results for different meshes are given in Figure 2 for the glasses mesh and in Figure 3 for the armadillo mesh. For the both meshes, the algorithm correctly finds the symmetry axis.

4. Pairwise Harmonics

In this part, I use pairwise harmonics for intrinsic reflectional symmetry axis detection. On a mesh surface, a harmonic function f defines a scalar field that satisfies the Laplace equation $\Delta f = 0$, subject to certain Dirichlet boundary constraints. Given a pair of points (p, q) on the mesh surface, the pairwise harmonic for (p, q) , which we denote as $f_{p,q}$, is defined as the harmonic field with Dirichlet boundary constraints $f(p) = 0$ and $f(q) = 1$. The discretization leads to a sparse linear system $Lf = b$, with L denoting the Laplacian matrix.

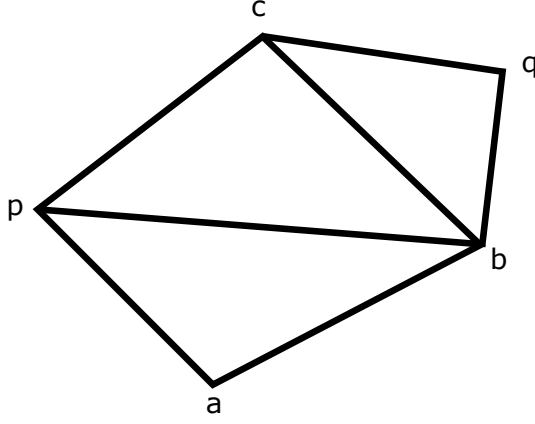


Figure 4: A mesh with 3 triangles and 5 vertices

I will describe how to form system on a simple mesh given in Figure 4. The mesh composed of 3 triangles and 5 vertices. The points p and q are boundary points with $f(p) = f_p = 0$ and $f(q) = f_q = 1$ and for points a, b, and c, the equations for uniform Laplacian are

$$\begin{aligned} 2f_a - f_b - f_p &= 0 \\ 3f_b - f_a - f_c - f_p - f_q &= 0 \\ 3f_c - f_b - f_p - f_q &= 0 \end{aligned} \tag{2}$$

Arranging this system of equations in matrix form after imposing the boundary conditions $f_p = 0$ and $f_q = 1$, we get

$$\begin{bmatrix} 2 & -1 & 0 \\ -1 & 3 & -1 \\ 0 & -1 & 3 \end{bmatrix} \begin{bmatrix} f_a \\ f_b \\ f_c \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \tag{3}$$

The matrix in the system is the Laplacian matrix of the mesh except the boundary points. The boundary conditions are present in the vector b .

For our case, I also used Uniform Laplacian. When I tried to use Cotangent Laplacian, I get meaningless results due to negative weights led by the mesh. Even if I turn the negative weights into 0, the results did not improve.

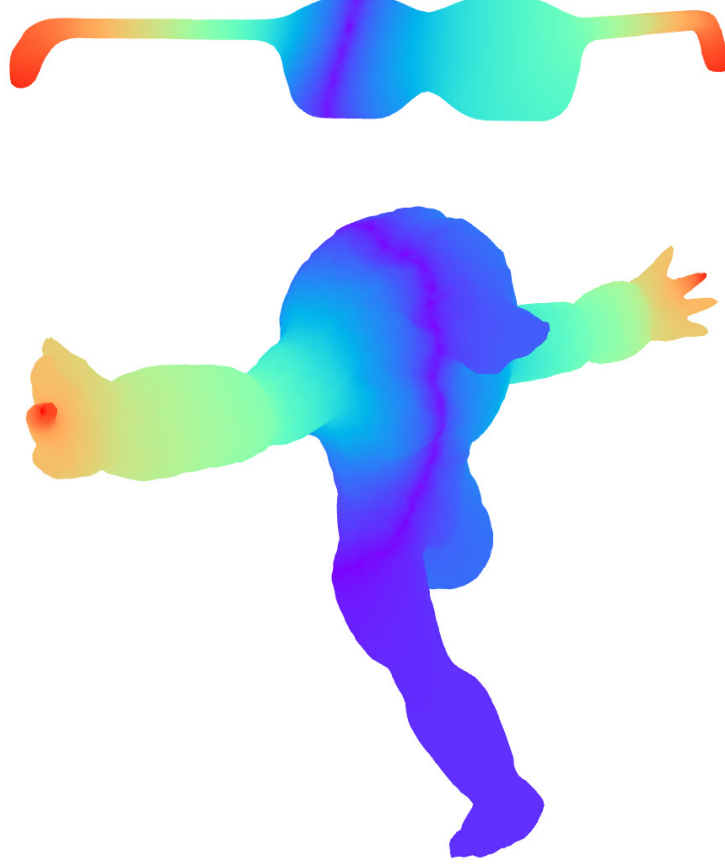


Figure 5: The initial results by harmonics analysis

The results for different meshes are given in Figure 5. The found axes are different than intrinsic symmetry axes. I believe that the mesh being non-uniform and not perfectly symmetric affects the results. For this reason, I tried to increase the boundary points. I found distances from each extremity to every vertex on the mesh and stored them in two arrays. The indices of the arrays represent each vertex on the mesh. I will use these two arrays to found additional boundary points.

Let V is a set composed of vertices of the mesh. For each extremities, let the distance arrays be δ_0 and δ_1 and boundary points sets be $B_0 \subset V$ and $B_1 \subset V$ respectively. Then, B_0 and B_1 is defined as

$$\begin{aligned} B_0 &= \text{where } [(\delta_0 - \delta_1) > \psi] \\ B_1 &= \text{where } [(\delta_1 - \delta_0) > \psi] \end{aligned} \tag{4}$$

where ψ is a threshold value.

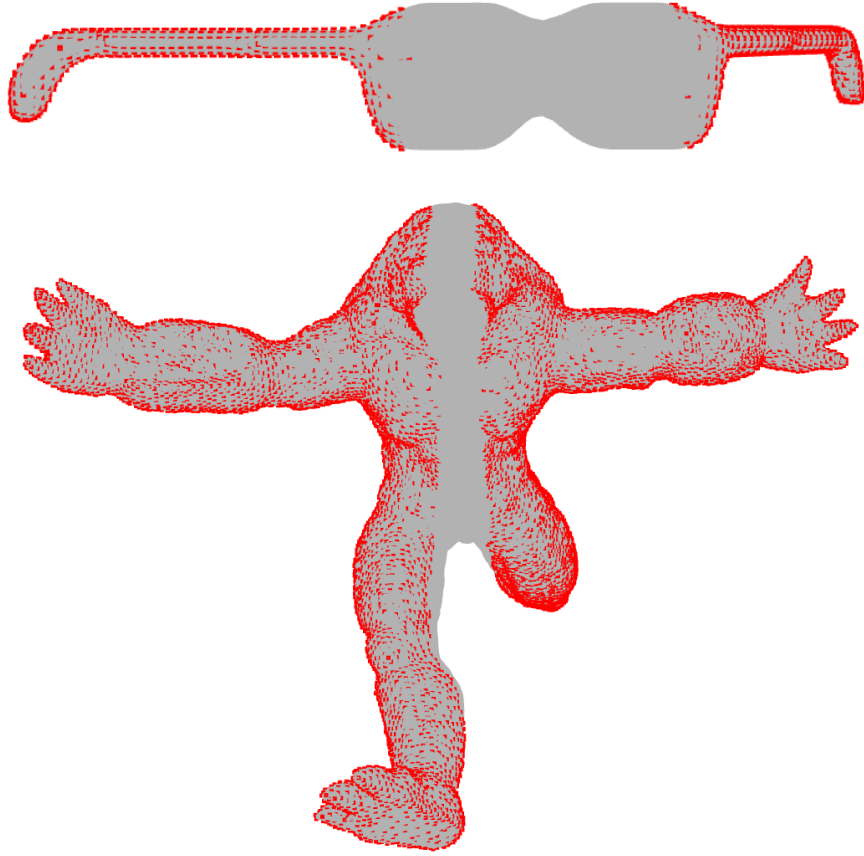


Figure 6: Increased number of boundary points by Equation (4) using $\psi = 0.5$ for the glasses mesh and $\psi = 0.1$ for the armadillo mesh

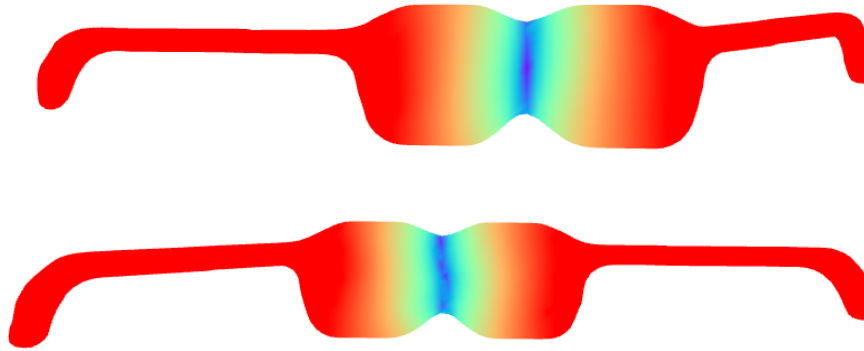


Figure 7: The results from different poses for the glasses mesh by harmonics analysis

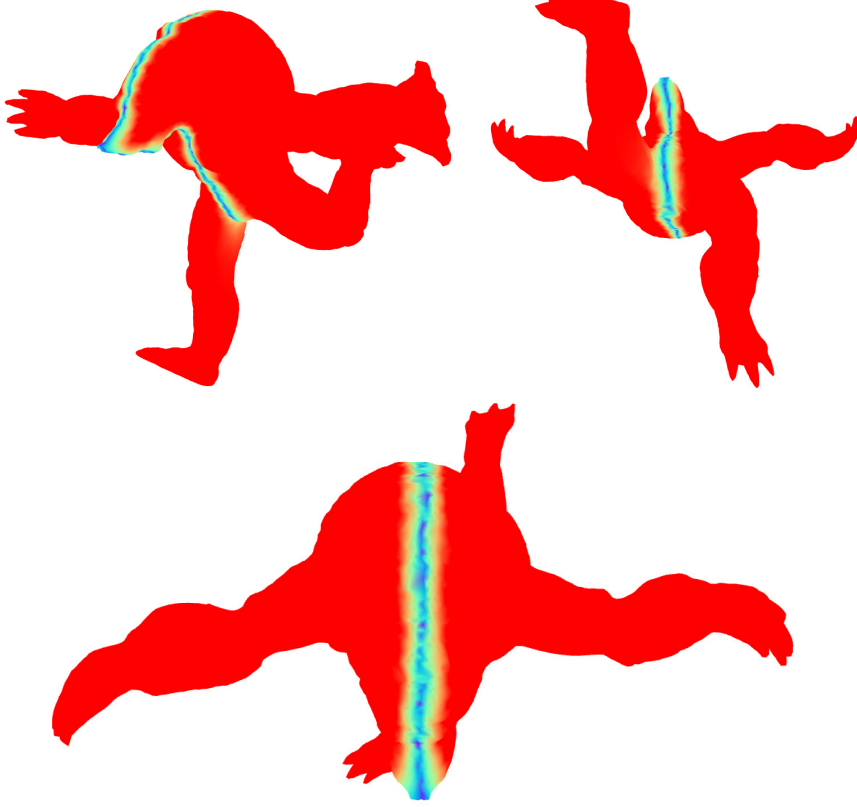


Figure 8: The results from different poses for the armadillo mesh by harmonics analysis

For the armadillo mesh, I used a threshold value $\psi = 0.5$ for the glasses mesh, and $\psi = 0.1$ for the armadillo mesh. This threshold value can also be arranged by the element size. Extracted boundary points for both meshes are depicted in Figure 6. By solving the system with these boundary points, I get the result given in Figure 7 for the glasses mesh and 8 for the armadillo mesh. Now we see that the result is much improved and we get a meaningful intrinsic symmetry axes.

5. Conclusions

In this assignment, we implemented and analyzed two distinct methods Geodesic Isocurve Intersection and Pairwise Harmonics Analysis for detecting the intrinsic symmetry axis of 3D shapes. Each approach contributed complementary strengths: the Geodesic Isocurve Intersection provided a straightforward geometric intuition, while the Pairwise Harmonics method offered a more robust, mathematically grounded perspective. Through the use of Farthest Point Sampling, geodesic distance computation, and similarity testing, we established a pipeline for identifying intrinsically symmetric point pairs that serve as critical input for both techniques.

Despite achieving accurate results on relatively simple or well-structured meshes (e.g., the glasses model), both methods encountered limitations when applied to complex, non-uniform, or less symmetrically shaped meshes. These challenges result from local geometric distortions, varying curvature, and the extent of the symmetry axis. Enhancements such as extending boundary point sets improved the results considerably.

Overall, this study highlights the potential of intrinsic symmetry detection in 3D meshes. While the implemented methods are effective under certain conditions,

further refinement—possibly involving adaptive weighting, or more sophisticated mesh processing techniques—would be necessary to handle a wider variety of shapes with higher reliability.

References

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